

Rakesh Adak

Backend Engineer

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Skills

Languages: C, C++, JavaScript, TypeScript, Python

Backend: Node.js, Express.js, Hono, RESTful API Design, WebSockets, Microservices

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (S3, Lambda), Docker, Cloudflare Workers, GitHub Actions (CI/CD)

Testing & Tools: Jest, Postman, Git

Education

Heritage Institute of Technology, Kolkata, B.Tech in Computer Science and Engineering 2024 – 2028
(Artificial Intelligence & Machine Learning)
CGPA: 8.00/10.00

Projects

W3Deploy — Serverless Deployment Platform July 2026

Hono, Zod, Ink, Cloudflare Workers, AWS S3 | [GitHub](#) | [NPM Package](#)

- Built a Vercel-style CLI for deploying and hosting static frontend applications, architecting a **100% serverless deployment pipeline** using Cloudflare Workers and D1 that reduced infrastructure costs by **90%+** compared to traditional VM hosting
- Reduced AWS S3 egress costs and achieved a **99%+ CDN cache hit ratio** by implementing an edge caching strategy with 1-year immutable TTLs via the Cloudflare Cache API
- Designed a presigned-URL upload architecture on AWS S3 supporting **1,000+ concurrent deployments**, and built an edge routing layer with Cloudflare Workers and Hono delivering **sub-50 ms global content delivery latency**

Nanobond — Unified Sovereign Bond Platform Jan 2026

Node.js, Express.js, TypeScript, Ethers.js, MongoDB, LayerZero | [GitHub](#) | [Visit](#)

- Engineered the backend for a unified platform enabling **fractional, cross-chain investment in sovereign government bonds**, reducing investment gas costs via an Account Abstraction (AA) architecture across **5 major EVM networks** (Ethereum, Arbitrum, Polygon, Scroll, Mantle)
- Accelerated cross-chain bond settlement to **under 10 minutes** by architecting an interoperability layer via LayerZero gateways, routing liquidity seamlessly from fragmented networks into Mantle
- Ensured **100% on-chain debt collateralization** and zero-delay payouts via yield bot and debt monitor services, while ensuring atomic transaction execution and eliminating double-spend risk through bundled smart contract operations

Achievements

Finalist, EIBS 2.0 (Kshitij, IIT Kharagpur) Jan 2026

Built and pitched Nanobonds, a fractional government bond platform, at one of India's largest techno-management fests

2nd Runner-Up, Dakksh26 Vibeathon Mar 2026

Placed 3rd overall at the college techfest for developing and pitching AgentX, an MCP server for crypto payments (x402)